B T Senthan Amuthan

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Professional Summary

Computer Science undergraduate with experience building production backend systems and scalable full-stack applications. Skilled in large codebases, distributed backend development, code reviews, testing, and cross-functional collaboration.

Education

Shiv Nadar University Chennai Aug 2023 – Aug 2027
B.Tech in Computer Science (Cyber Security) ***CGPA: 9.26*** *Kalavakkam, India*

Experience

Summer Analyst Intern – Citi June 2026 – July 2026
Software Engineering *India*

- Built the backend foundation of an internal enterprise platform, **designing REST APIs**, database schemas, and migration workflows for production use.
- Designed and implemented **authentication middleware** with structured logging, request tracing, and authorization, securing **100%** of protected APIs while improving debugging across backend services.
- Resolved recurring **CI/CD** failures through log and monitoring analysis, restoring **100% pipeline reliability**, achieving a **99%** code-quality gate pass rate, and reducing test execution time by **10×** using isolated test databases with **250+ automated tests** and **100% coverage**.
- Quickly ramped up on a **large enterprise codebase** by collaborating with senior engineers, participating in design discussions, and incorporating review feedback to deliver production-ready backend features.
- Delivered **4 production stories** through cross-functional collaboration, contributing **6+ pull requests** with **zero production defects** while participating in design discussions and code reviews.

Software Development Engineer Intern June 2025 – August 2025
Dexio Designs (stipend internship) *India*

- Designed **HLD and LLD** for production-ready full-stack features before implementing them, collaborating with Product Owners from design through deployment.
- Developed **reusable UI components** and REST API integrations, reducing QA revisions by **50%** while improving maintainability and delivery speed.

Projects

AstraStore – Distributed Storage Engine *Java, Linux, Streaming I/O, REST/gRPC, Docker*

- Engineered a low-level storage node agent with a **zero-memory streaming chunker** (dual SHA-256 digests) and atomic **fsync-validated writes**, guaranteeing **100% data integrity** with near-zero memory overhead during node failures.
- Built an automated background self-healing scanner (**UnderReplicationScanner**) and peer-to-peer replication service with exponential backoff, reducing degraded chunk recovery time by **65%**.

QEMU Lab Orchestrator *QEMU, Express.js, Docker, Guacamole, Next.js*

- Designed a **virtualization platform** orchestrating isolated QEMU VMs via REST APIs, integrating Dockerized **Apache Guacamole** for low-latency browser access under **50ms latency**.
- Automated end-to-end VM lifecycle management and persistent state tracking, cutting VM provisioning time by **80%** across multi-tenant lab topologies.

Open Source Contributions

Bento Slides – Ghost Cut Support *Merged PR #679*

- Delivered **Ghost Cut support** for slide elements, enabling users to move selections across slides through keyboard and clipboard interactions, with cancellation, restoration, and single-step undo.
- Integrated ghost rendering with model-derived opacity and browser-local preferences; validated the feature through successful builds, translation checks, and manual interaction testing before upstream merge.

Skills

- **Languages:** C++, Java, Python, JavaScript, TypeScript, Rust, C
- **Backend:** FastAPI, Node.js, Express.js, REST APIs, Microservices, WebSockets, Async Programming
- **Databases:** PostgreSQL, MongoDB, SQLite, SQL, Prisma, Liquibase
- **DevOps:** Git, Docker, Kubernetes, OpenShift, CI/CD, IBM S3, Linux, SonarQube, Grafana
- **Software Engineering:** DSA, OOP, SDLC, System Design, Unit Testing, Code Reviews

Achievements

- Solved **400+** DSA problems on LeetCode; active participant in competitive programming and technical hackathons including **SIH 2023 & 2024** and **SNUHacks**.
- Named **Finalist** at Envithon hackathon; completed technical workshops on **DevSecOps** and secure CI/CD pipeline automation.